

Advanced Java Programming (3-0-3) (Elective)

Evaluation:

	Theory	Practical	Total
Sessional	30	20	50
Final	50	-	50
Total	80	20	100

Objectives:

1. To enable the students to understand the principles of the Java Language.
2. To enable students to learn to produce well designed, dynamic web applications.
3. To introduce tools, technologies and framework hence Java Beans, Servlets, JSP and EJB are introduced to enhance web development skills.

Course Contents:

- 1. Multimedia: Applets and Application** **5hrs**
 - 1.1 Introduction to multimedia
 - 1.2 Loading multimedia components using JMF (image, audio, videos)
- 2. RMI (Remote Method Invocation)** **5hrs**
 - 2.1 Introduction and Architecture of RMI
 - 2.2 Java RMI classes and Interfaces
 - 2.3 Writing simple RMI application
 - 2.4 Parameter passing in remote methods (marshalling and unmarshalling)
 - 2.5 Introduction to CORBA
- 3. JSP: (Java Server Pages)** **8hrs**
 - 3.1 Introduction and lifecycle of JSP
 - 3.2 JSP Structure
 - 3.3 Writing JSP Program
 - 3.4 JSP Actions: include, forward, and plugin, Managing session using JSP, JSP and Database
 - 3.5 Error Handling in JSP
 - 3.6 Working with Java Mail-Understanding Protocol in Java mail components Java mail
- 4. Java Beans** **8hrs**
 - 4.1 Brief Introduction to JEE, Java Bean Introduction, design pattern
 - 4.2 Writing simple bean
 - 4.3 Bean persistence and introspection
 - 4.4 EJB
 - 4.4.1 Architecture
 - 4.4.2 Container classes, Interfaces
 - 4.4.3 EJB types- Session, Entity, Message Driven
- 5. Struts2** **10hrs**
 - 5.1 Introduction to MVC Architecture
 - 5.2 Struts architecture and environmental setup
 - 5.3 Struts configuration
 - 5.4 Struts actions

- 5.5 Struts interceptors
- 5.6 Execute Struts application in NetBeans
- 5.7 Struts tags –control tags, form tags, data tags, and AJAX tags
- 5.8 Struts value stack/OGNL
- 5.9 Struts Validations
- 5.10 Struts Exception handling

6. Introduction to Hibernate

7hrs

- 6.1 Overview of Hibernate
- 6.2 Hibernate Architecture
- 6.3 Understanding Hibernate <generator> element
- 6.4 Understanding Hibernate ORM
- 6.5 Struts2 Integration with Hibernate

7. Internationalization

5hrs

- 7.1 Introduction to internationalization
- 7.2 Java support for i18n
- 7.3 Providing localized resources
- 7.4 Working with dates, numbers, and currencies

Text Books:

1. Herbert Schildt, Java The Complete Reference, Tata McGraw Hill Edition

Reference Books:

1. Hans Bergsten- Java Server Pages, Publisher O'Reilly
2. Dave Minter and Jeff Linwood- Beginning Hibernate from novice to professional, A press
3. Kogent, Java 6 Programming Black Book, Dreamtech Publications
4. E. Balaguruswami, Programming with Java, A Primer, Tata McGraw Hill Edition